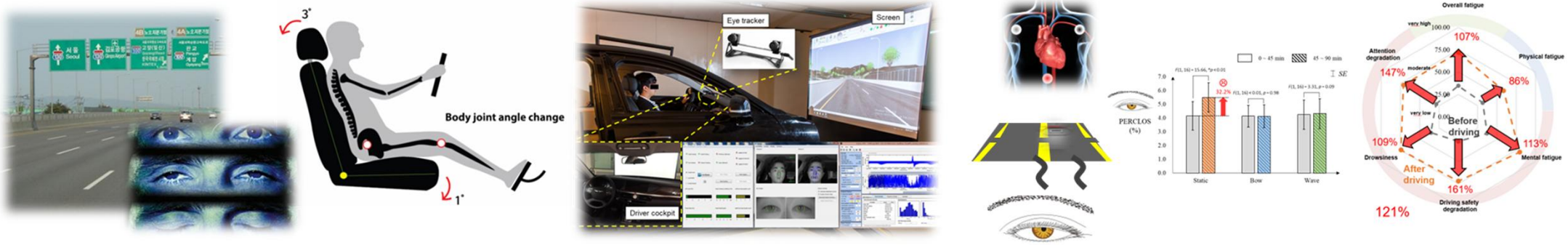
Research Methodologies
in Human Factors & Ergonomics (HF/E)



The effects of a motion seat system on driver's passive task-related fatigue: A lab-based driving simulation study (You et al., 2018; sponsored by Hyundai Motor Group)

Contents

Part I

- Introduction to HF/E Research Methods
- Three Types of HF/E Research
- Field vs. Laboratory Settings
- Variables & Control of Confounds
- Participant Sampling & Power

Part II

- Human Criteria
- Psychometric Requirements of Criteria
- Practical Requirements of Criteria
- Statistical Techniques
- Integration & Applications
- Quiz

Learning Goals (1/2)

GOALS

- Distinguish among **system-descriptive criteria, task performance criteria, and human criteria**.
- Compare **objective measures with subjective measures** in HF/E research and explain the **scientific benefit of using both measures simultaneously** to assess human performance.
- Identify **performance measures, physiological measures, behavioral measures, and subjective measures** used to indicate human physical or mental states.
- Explain the **psychometric requirements** for human criteria considered in HF/E research: **reliability, validity, free of contamination, and sensitivity**.
- Distinguish among **face validity, content validity, and construct validity**.

- Describe the **practical requirements** for human criteria considered in HF/E research: **objectivity, quantitativeness, unobtrusiveness, ease of use, speed, and cost**.
- Distinguish between **descriptive statistics and inferential statistics**.
- Select appropriate **statistical techniques (e.g., correlation, t-test, ANOVA)** based on specific experimental designs.

Learning Goals (2/2)

GOALS

- Compare **statistical significance (*p*-value)** with **practical significance (effect size)** in interpreting findings.
- Synthesize **objective and subjective measures** to comprehensively evaluate a human-machine system.
- **Apply HF/E research methods and criteria** to evaluate a real-world ergonomic solution.

Types of Criterion Measures

Category	System-Descriptive Criteria	Task Performance Criteria	Human Criteria
Focus	**Engineering & System Attributes**	**Outcome of System + Human**	**Behaviors & Internal Responses**
Definition	Criteria describing the functioning or characteristics of the entire system (hardware/software), often independent of specific user actions.	Criteria reflecting the output or result of a task where the human is a component; often more global than human-specific measures.	Criteria dealing specifically with the human's responses, biological state, and subjective experiences during performance.
Key Dimensions	• Reliability (MTBF) • Maintainability • Safety features • Operational costs	• Quantity (Output) • Quality (Accuracy) • Time (Speed/Duration)	• Performance measures • Physiological & behavioral measures • Subjective measures
Examples	• Maximum RPM • Weight • Radio interference • Resistance to wear	• Tons moved • Errors • Delay time	• Reaction time • Heart rate • NASA-TLX ratings

Types of Human Criteria

Category	Objective Measures	Subjective Measures
Definition	Direct measurement of observable behavior, physiological states, or system performance independent of observer bias.	Self-reports of internal states, opinions, preferences, or perceived experiences based on user introspection.
Key Types	**1. Performance:** Speed, Accuracy, Throughput **2. Physiological:** HR/HRV, EDA, EMG, EEG, Eye-tracking **3. Behavioral:** Gaze patterns, movement kinematics	**1. Rating Scales:** Unidimensional (Likert) vs Multidimensional **2. Verbal Protocols:** Think-aloud (concurrent/retrospective) **3. Questionnaires:** Interviews, Focus groups
Examples	• Reaction time • Task completion time • Error rate • Heat rate variability • Fixation duration	• NASA-TLX (workload) • SUS (usability) • Borg RPE (exertion) • Preference • Post-task interview
Advantages	✅ High reliability and replicability ✅ Quantitative and verifiable (Hard data) ✅ Often unobtrusive (e.g., system logs)	✅ Direct access to user experience (e.g., "why" it happened) ✅ Easy and inexpensive to administer ✅ High face validity to stakeholders
Disadvantages	❌ Does not explain "why" (no diagnostic insight) ❌ Physiological measures can be intrusive/costly	❌ Susceptible to memory bias and social desirability ❌ Dissociation from performance (e.g., happy but failing)
When to Use	When precise measurement of human capabilities, limits, or system efficiency is required. Essential for safety-critical systems.	When assessing user acceptance, preference, perceived workload, or diagnosing reasons for errors. Essential for consumer products.

Objective Measures: Performance Measures

Speed / Efficiency

Measures the temporal aspect of performance.

- Task completion time, reaction time (RT)
- Units processed per hour
- Movement time (Fitts' Law applications)

Accuracy / Errors

Measures the correctness or quality of performance.

- Error rate (%), number of errors
- Omission errors vs. Commission errors
- Slip vs. Mistake classification

Process Measures

Reveals *how* the task was performed, not just the outcome.

- Keystrokes, mouse clicks, path efficiency
- Control inputs, dwell times on areas of interest

Throughput

A composite measure combining speed and accuracy, often derived from information theory (bits/sec).

Throughput = Index of Difficulty / Movement Time

Evaluation of Objective Measures

Pros

- Directly observable & verifiable
- High face validity for management
- Quantitative & statistical power
- Often non-intrusive (logging)

Cons

- Doesn't explain "why" (cognitive)
- Can miss high cognitive cost (workload)
- Speed-Accuracy Trade-off (SATO) complexities

Objective Measures: Physiological & Behavioral Measures

Objective indices of internal state (arousal, stress, cognitive load) and overt behavior patterns.

Cardiovascular (HR/HRV)

Heart Rate (HR) and Variability (HRV) measure mental workload and stress responses. Decreased HRV often indicates higher cognitive load.

Electrodermal (EDA/GSR)

Measures skin conductance changes due to sweat gland activity. Indicative of emotional arousal, anxiety, and sympathetic nervous system activation.

Brain Activity (EEG)

Measures electrical brain waves. Alpha waves correlate with relaxation; Theta/Beta ratios indicate engagement or fatigue levels.

Eye-Tracking

Fixations (attention focus), saccades (search efficiency), and pupil diameter (cognitive load). reveals "where" and "how" users process visual info.

Muscle Activity (EMG)

Electromyography measures muscle tension. Used for physical workload, fatigue assessment, and even detecting subtle facial expressions.

Behavioral Kinematics

Motion capture, dwell times, and movement paths. Quantifies efficiency of physical interaction and ergonomic posture risks.

Trade-offs: Pros & Cons

Advantages

- Continuous measurement (real-time data)
- Highly sensitive to subtle changes in state
- Objective (harder for participants to fake)
- Can measure unconscious responses

Disadvantages

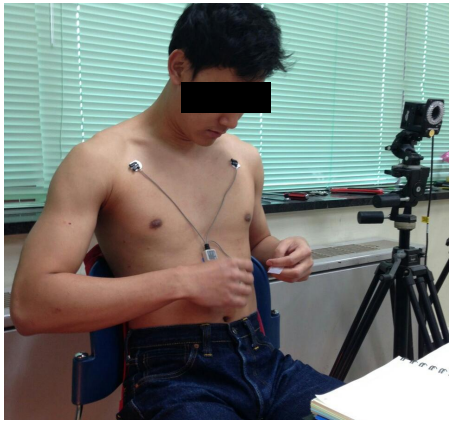
- Often intrusive (sensors, wires, constraints)
- Requires complex calibration and expertise
- Data noisy; prone to movement artifacts
- "Reverse Inference" problem (ambiguous meaning)

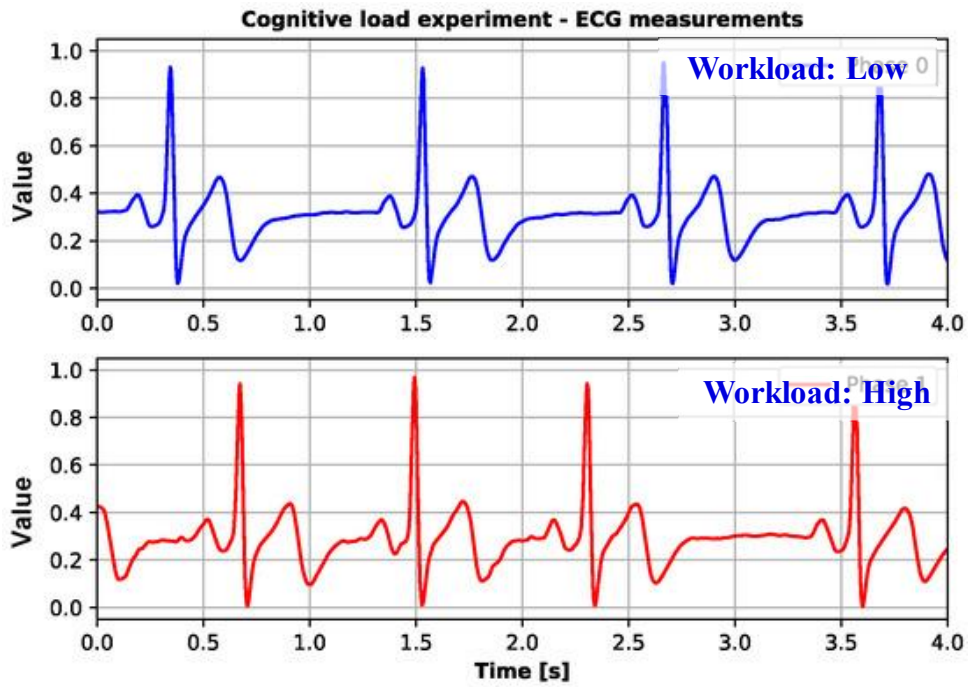
Corrected

HF/E Tool: ECG

An electrocardiogram (ECG or EKG) is a non-invasive, quick, and painless test that **records the heart's electrical activity** using electrodes attached to the skin.

Noraxon Telemyo DTS
(Noraxon Inc., USA)

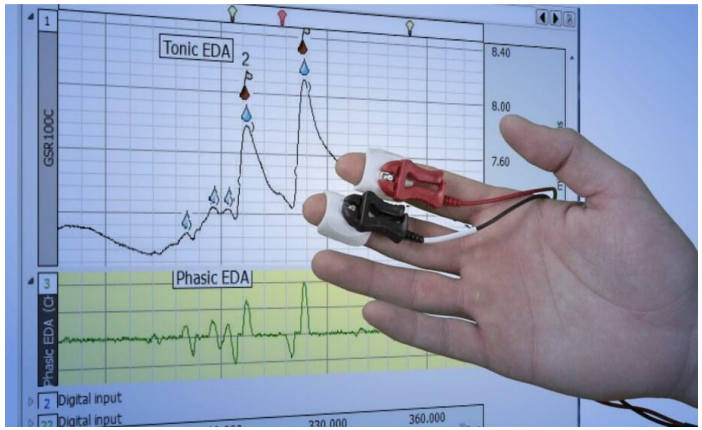


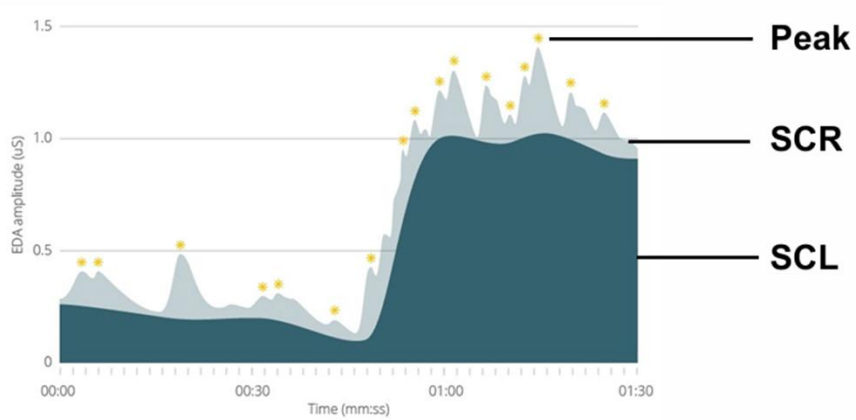


HF/E Tool: EDA/SCR

Electrodermal Activity (EDA), also known as skin conductance response (SCR) or Galvanic Skin Response (GSR), **measures changes in the skin's electrical conductivity r**esulting from eccrine sweat gland activity, which is directly regulated by **the sympathetic nervous system**.

Flexcomp Pro
(Thought Technology, Canada)

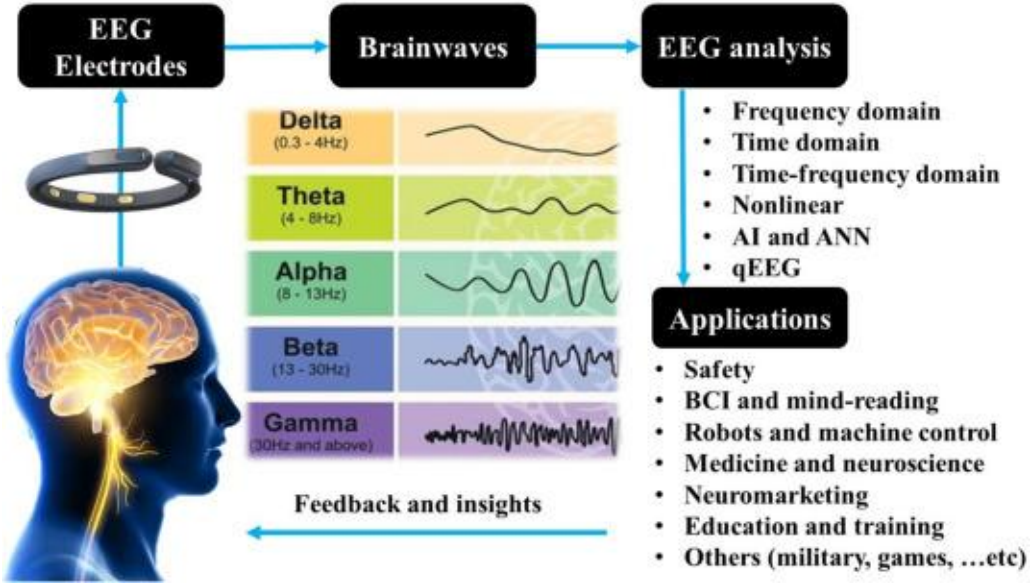




Corrected

HF/E Tool: EEG

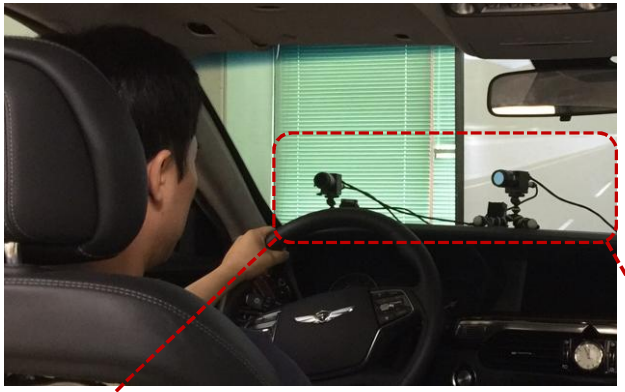
An EEG system (Electroencephalogram System) is a non-invasive medical or research tool used to **monitor, record, and analyze the electrical activity of the brain**.

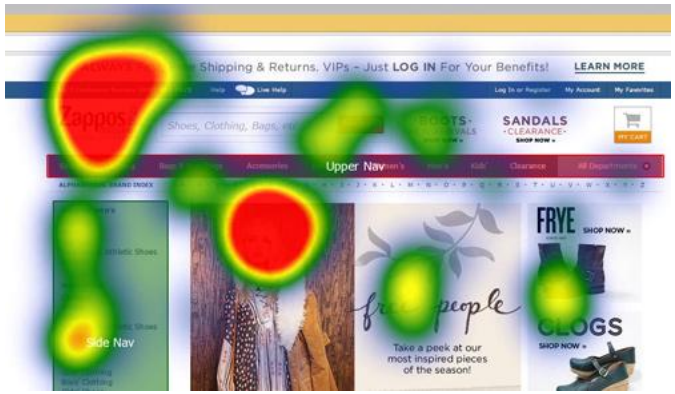


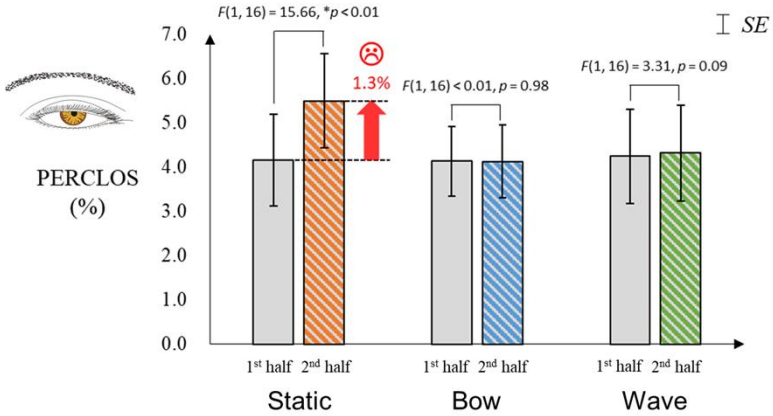
HF/E Tool: Eye Tracking

Eye tracking is the process of measuring either **the point of gaze (where one is looking) or the motion of an eye relative to the head**, typically using specialized sensors or cameras.

faceLAB 5
(Seeing Machines, Australia)







Subjective Measures: Ratings & Reports

Standardized Scales

Mental Workload (NASA-TLX)

Multidimensional rating scale with 6 subscales:

> Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, Frustration

Usability (SUS)

System Usability Scale: Quick, 10-item scale (Likert) for assessing global subjective usability.

> Scores 0-100; Industry standard for "quick and dirty" assessment

Discomfort & Pain

Borg RPE (Rating of Perceived Exertion), Corlett & Bishop Body Map.

Qualitative Methods & Evaluation

Verbal Protocols

Think-Aloud Protocol (concurrent or retrospective).

> Reveals cognitive processes, strategies, and misconceptions not visible in performance data

Trade-offs: Pros & Cons

✓ Advantages

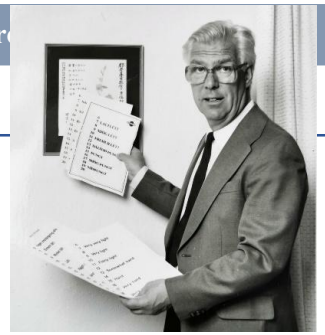
- Captures user's perspective
- Cheap
- Easy to administer
- Face validity

× Disadvantages

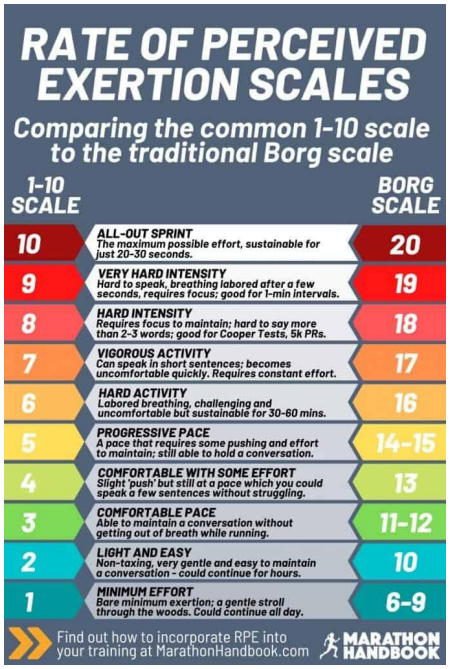
- Memory bias
- Inaccurate
- Dissociation from objective performance

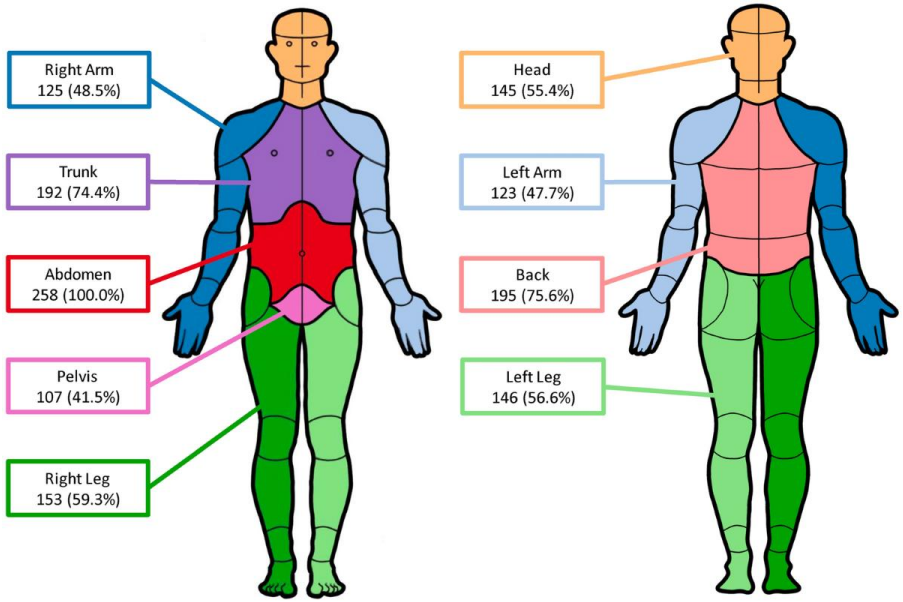
HF/E Tool: Borg RPE & Body Map

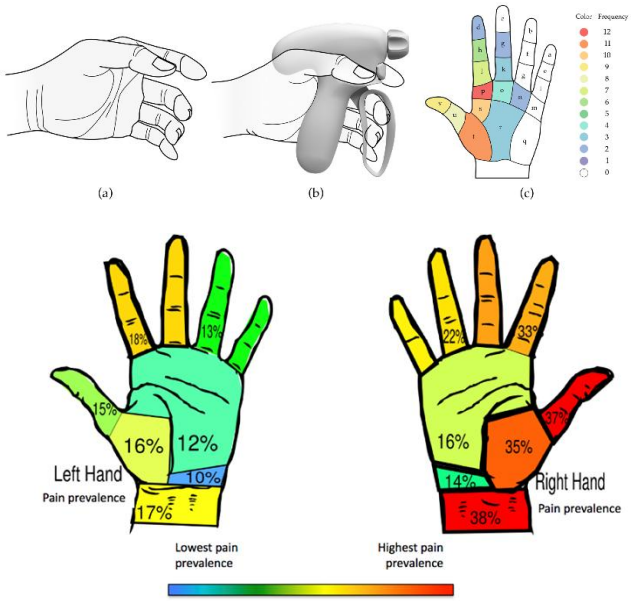
The Borg RPE (Rating of Perceived Exertion) scale is a **subjective tool** used to **measure physical exertion based on how hard you feel** your body is working, which is highly **correlated with heart rate**.



Gunnar Borg, PhD, MD. (1927-2020)
Professor emeritus of Perception
and Psychophysics
at Stockholm University







Triangulation: Multi-Method Measurement

Converging on the Truth

Comprehensive Integration

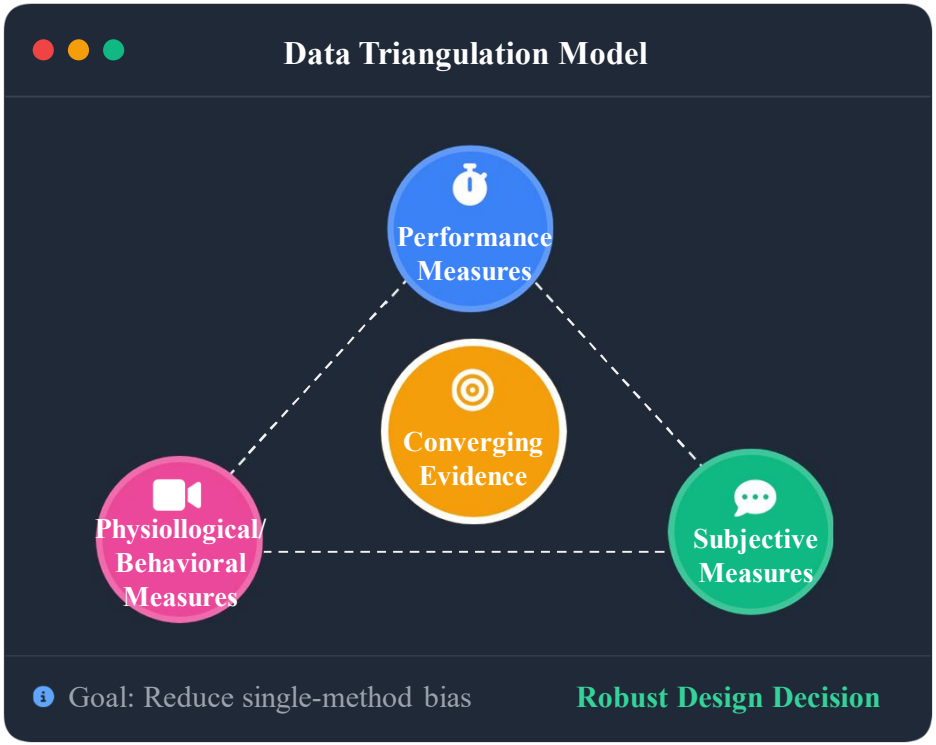
Combining performance data, physiological & behavioral observations, and subjective ratings to build a holistic picture of the human-system interaction.

Cross-Validation

Using one method to validate another. For example, does high physiological arousal (HRV) correlate with high subjective workload ratings (NASA-TLX)?

Resolving Discrepancies

Discrepancies (e.g., fast performance but high frustration) reveal hidden design trade-offs that single measures might miss.



Psychometric Requirements of Criteria

Reliability

Consistency or stability of measurement over time or across raters. Freedom from unsystematic random error.

Assessment Methods

- Test-Retest (Stability over time)
- Internal Consistency (Cronbach's α)
- Inter-Rater (Agreement between observers)

How to Improve

- Standardize instructions & environment
- Increase number of items/trials
- Train observers rigorously

Validity

The extent to which a variable actually measures what it is intended to measure (accuracy).

Types of Validity

- Face: Looks appropriate to users
- Content: Covers domain adequately
- Construct: Taps underlying theoretical trait

Evidence Needed

- Convergent: Correlates with similar measures
- Discriminant: Distinct from other traits
- Predictive: Predicts future performance

Freedom from Contamination

Measure should not be influenced by extraneous variables unrelated to the construct of interest.

Common Sources

- Practice/Learning effects
- Fatigue or Boredom
- Instrumentation drift/Calibration loss
- Environmental noise/distractions

Minimization Strategies

- Counterbalancing conditions
- Frequent calibration checks
- Constant environmental controls

Sensitivity

Ability of the measure to detect meaningful differences or changes in the variable of interest.

Issues to Avoid

- Ceiling Effect: Task too easy, scores max out
- Floor Effect: Task too hard, scores bottom out
- Coarse granularity (e.g., Pass/Fail only)

How to Increase

- Use continuous scales vs discrete categories
- Increase task difficulty to avoid ceiling
- Use multiple measures (Triangulation)

Reliability

*Reliability refers to the degree to which **a measure produces stable and consistent results**.*

Test-Retest Reliability

Measures stability over time. Obtained by administering the same test twice to the same group of participants after a certain time interval.

→ Critical for measuring stable traits (e.g., personality, intelligence).

Inter-Rater Reliability

Assess the degree to which different raters/observers give consistent estimates of the same phenomenon.

→ Essential for subjective ratings (e.g., posture analysis, verbal protocols).

Internal Consistency

Assess the consistency of results across items within a test. Do all items measure the same underlying construct?

→ Used for questionnaires (e.g., SUS, NASA-TLX).

Cronbach's Alpha (α)

The most common measure of internal consistency reliability.

Rules of Thumb

$\alpha \geq 0.9$	Excellent
$0.8 \leq \alpha < 0.9$	Good
$0.7 \leq \alpha < 0.8$	Acceptable
$\alpha < 0.7$	Questionable/Poor

Note: High alpha doesn't guarantee unidimensionality.

Validity

Validity refers to the degree to which a measure ***accurately assesses the specific concept it is intended to measure****.*

Face Validity

- Does the measure **appear** to be appropriate to test-takers and stakeholders?
- Subjective judgment, not statistical proof.
- Critical for participant motivation and acceptance (e.g., flight simulator realism).

Content Validity

- The extent to which the measure covers the **entire domain** of the construct.
- Established by expert review and defining the "universe" of content.
- *Example:* A usability test must assess all key tasks, not just the easiest one.

Construct Validity

The degree to which a test measures the theoretical construct it claims to measure. The "gold standard" of validity.

Convergent Validity

The measure correlates strongly with other established measures of the same construct.

e.g., New workload scale scores correlate with NASA-TLX.

Discriminant Validity

The measure does **not** correlate with measures of unrelated constructs.

e.g., Workload scores should not just reflect physical fatigue.

"Validity is not a property of the test itself, but of the interpretation of the test scores."

Freedom from Contamination

> *Freedom from contamination refers to the degree to which* ***a measure is free from extraneous variables or factors that are irrelevant to the construct it intends to assess****.*

Sources of Contamination

Practice & Fatigue Effects

Performance changes driven by repetition or exhaustion rather than the experimental manipulation.

Reactivity (Hawthorne Effect)

Participants modify their behavior simply because they know they are being observed or evaluated.

Instrumentation Drift

Changes in sensor calibration or observer rating criteria over the course of the study.

Environmental Noise

External distractions (lighting, sound, temperature) that introduce irrelevant variance into the measure.

Minimization Strategies

Counterbalancing & Randomization

Distribute order effects evenly across conditions to neutralize practice and fatigue confounds.

Unobtrusive Measurement

Use sensors, one-way mirrors, or naturalistic observation to minimize participant awareness.

Strict Standardization

Maintain rigorous protocols for instructions, environment, and equipment calibration.

Blinding Techniques

Keep participants (single-blind) and raters (double-blind) unaware of conditions to reduce bias.

Contamination occurs when the measure reflects constructs other than the one intended. A valid criterion must be "pure" from these extraneous influences.

Sensitivity

> *Sensitivity refers to the degree to which* ***a measure can detect small but meaningful changes or differences in the variable it is designed to assess****.*

Detecting Differences

The capability of an instrument to distinguish small but meaningful differences in the attribute being measured. A sensitive measure can detect subtle changes in workload or performance.

Scale Granularity

Sensitivity is linked to scale resolution. Coarse measures (e.g., binary Pass/Fail) lose information compared to fine-grained continuous scales (e.g., Response time in ms).

Impact on Statistical Power

Low sensitivity increases error variance, reducing the statistical power to detect an effect. Insensitive measures may lead to Type II errors (failing to reject a false null hypothesis).

Ceiling Effect

Occurs when a task is too easy or a scale range is too low. Scores cluster at the maximum (e.g., 99% accuracy), masking differences between high-performing conditions.

Floor Effect

Occurs when a task is too difficult. Scores cluster at the minimum (e.g., near 0 performance), failing to differentiate between low-performing conditions.

Mitigation Strategy

Conduct **pilot testing** to calibrate task difficulty and ensure the measurement scale covers the full range of expected behaviors without hitting boundaries.

Example: Knights of the Roundtable

Assessing "Combat Skill"

Goal: Determine **the best knights** for King Arthur's army

The Test Protocol:

Each knight **shoots one arrow at a stationary target**. **Distance from the center** is measured to **the nearest yard**.

Key Takeaway

A seemingly simple test can fail on multiple psychometric aspects if not carefully designed.

Reliability Problem

Issue: Shooting just one arrow means *scores vary wildly* day-to-day (luck vs. skill).

Fix: Average score of 10 arrows per day.

Validity Problem

Issue: Stationary targets don't represent *actual combat skill* (construct validity).

Fix: Include moving targets or sparring.

Contamination Problem

Issue: *Wind gusts or bent arrows affect scores* but aren't part of "skill".

Fix: Test indoors with standardized equipment.

Sensitivity Problem

Issue: *Measuring to the "nearest yard" masks subtle differences* between elite archers.

Fix: Measure distance in inches or centimeters.

Practical Requirements of Criteria

*"A good criterion must balance **scientific rigor (validity, reliability)** with **practical feasibility (cost, time, ease of use)**."*

01 Objectivity
Measurement is independent of observer bias; two different observers should obtain the same results.

02 Quantitativeness
Results are numerical and scalable rather than purely descriptive, allowing for statistical analysis and comparison.

03 Unobtrusiveness
The measurement process minimally interferes with the task, avoiding "reactivity" where observation alters behavior.

04 Ease of Use
Simple collection procedures requiring minimal specialized training or complex instrumentation setup.

05 Speed (Quickness)
Data collection and analysis can be completed rapidly to provide timely feedback in the iterative design cycle.

06 Inexpensiveness
Low financial cost for equipment, software, and personnel time, making the method feasible within project budgets.

Applying Practical Criteria: Selection Matrix

Evaluate candidate measures against practical constraints to select the optimal set for your study.

Good/Desirable Moderate/Neutral Poor/Undesirable

Criterion	Objectivity	Quantitative-ness	Unobtrusiveness	Ease of Use	Speed / Cost	Typical Application
Task Time/Errors (Objective Performance)	High	High	High	High	Low Cost	Usability Testing, Production lines
NASA-TLX (Subjective Rating)	Low (Self-report)	Med (Ordinal scale)	Med (Interrupts task)	High	Low Cost	Workload assessment, Design comparison
Eye-Tracking (Physiological)	High	High	Med/Low (Headgear/Chinrest)	Low (Calibration needed)	High Cost	Visual attention studies, Cockpit design
Verbal Protocol (Subjective Report)	Low	Low (Qualitative)	Low (Changes process)	Med	Med Cost (Analysis time)	Cognitive walkthrough, Expert troubleshooting
Heart Rate (HRV) (Physiological)	High	High	Med (Sensors attached)	Med	Med Cost	Stress monitoring, Fatigue studies

Statistical Techniques: Mapping to Research Type

Factor	Descriptive Research	Experimental Research	Evaluation Research
Analysis Goal	**Summarize & Associate** Identify patterns, distributions, and relationships between variables.	**Compare & Explain** Test causal effects of IVs on DVs; identify significant differences.	**Benchmark & Decide** Verify if system meets performance or usability standards.
Descriptive Statistics	• Mean, Median, Mode • SD, Variance, Range • Frequency counts, % • Percentiles (anthropometry)	• Group Means & SE • **Effect Size** (Cohen's d, η^2) • Confidence Intervals (95%)	• Success rates / Errors • Mean Satisfaction (SUS) • % of users meeting criteria • Completion time
Inferential Statistics	• Pearson Correlation (r^2) • Spearman Rank (ρ) • Chi-square (χ^2) • Simple Regression	• ***t*-tests** (Paired / Independent groups) • **ANOVA** (One-way, Factorial, Repeated Measures) • Post-hoc tests (Tukey, Bonferroni)	• One-sample *t*-test (vs. benchmark) • Binomial Test (pass/fail) • Wilcoxon Signed-Rank • Confidence Interval overlap
Visualizations	Histograms, Scatterplots, Pie charts, Heatmaps	Interaction plots, Box plots, Bar charts with Error Bars	Radar charts, Scorecards, Benchmark lines

Analysis Selection Tip:
Choose your test based on data type (Nominal/Ordinal vs Interval/Ratio) and study design (Within-subjects vs Between-subjects).

Example: Comparing workload (NASA-TLX) across 3 displays (Within-subjects) $\rightarrow$ **Repeated Measures ANOVA**

Correlation: Association, Not Causation

Definition & Purpose

Measures the strength and direction of a linear relationship between two variables. Often used in descriptive research to scope problems or model inputs.

Common Coefficients

Pearson's r: Parametric; assumes normal distribution and interval/ratio data.
Spearman's ρ: Non-parametric; uses rank order; robust to outliers.

Interpretation

Range: **-1.0** (Perfect Negative) to **+1.0** (Perfect Positive).
0 implies no *linear* relationship.

> Effect Size Benchmarks (Cohen):
> Small ($r = .10$), Medium ($r = .30$), Large ($r = .50$)

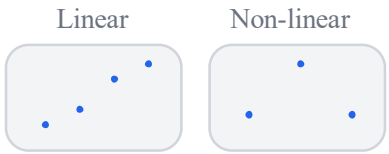
Coefficient of Determination R^2)

Represents the proportion of variance in one variable explained by the other.

> Example: If $r = 0.80$, then $R^2 = 0.64$ (64% variance explained).

The Necessity of Visual Inspection

Always inspect scatterplots before calculation to detect non-linearity (e.g., U-shaped Yerkes-Dodson law).



Limitations

Correlation ≠ Causation: Correlation does not imply directionality.
Third Variable Problem: A confound (Z) may cause both X and Y.

HF/E Example: Correlation between *Age* and *Reaction Time* allows prediction but does not prove age causes slowing (health or activity level could be confounds).

t-Tests: Comparing Means

Independent Samples *t*-test

Compares means of two separate/independent groups (Between-Subjects Design).

> ***HF/E Example:*** *Comparing reaction times between a group using a Joystick vs. a group using a Mouse.*

Paired Samples *t*-test

Compares means of the same group in two different conditions (Within-Subjects Design).

> ***HF/E Example:*** *Comparing workload (NASA-TLX) scores for Design A and Design B from the same participants.*

Key Assumptions

Data is continuous (Interval/Ratio); Normal distribution (Shapiro-Wilk); Homogeneity of variance (Levene's test for independence).

Effect Size (Cohen's *d*)

Measures the magnitude of the difference, independent of sample size. Crucial for practical significance.

Benchmarks: Small (0.2), Medium (0.5), Large (0.8).

Reporting Standards

Report the t-statistic, degrees of freedom (df), p-value, and effect size. Always include Mean (M) and SD.

> "Task completion time was significantly lower for the new interface (Mean = 14s, SD = 2.1) than the old interface (Mean = 18s, SD = 3.5), $t(29) = -4.12$, $p < .001$, $d = 1.34$."

Common Pitfalls

Using multiple *t*-tests for > 2 groups increases Type I error (use ANOVA instead). Ignoring normality violations.

ANOVA: Main Effects & Interactions

Analysis of Variance (ANOVA)

Extends t-tests to compare means across 3+ groups (One-way) or multiple independent variables (Factorial). Controls Type I error rates better than multiple t-tests.

Main Effects & Interactions

Main Effect: The independent impact of a single variable.
Interaction: When the effect of one variable depends on the level of another. Crucial for identifying design trade-offs.

Post-Hoc Tests

Required when the omnibus F-test is significant to determine exactly *which* groups differ (e.g., Tukey HSD, Bonferroni corrections).

HF/E Application: 2×2 Design

IVs: Display Modality (Visual vs. Auditory) × Workload (Low vs. High)
DV: Response Time

Interpreting the Interaction

An interaction might reveal that *Auditory Displays* are superior, but **only** under *High Workload* conditions. Main effects alone would mask this insight.

Assumptions to Check

Normality of residuals, homogeneity of variance (Levene's test), and sphericity (for repeated measures). Violations may require corrections (e.g., Greenhouse-Geisser).

Key Insight: In HF/E, interactions are often more practically significant than main effects because they define the boundary conditions for safe system operation.

Assumptions, Alternatives, and Reporting

1 Check Parametric Assumptions

Always verify assumptions before interpreting ANOVA/*t*-test results. If violated, Type I error rates may inflate.

- Normality (Shapiro-Wilk test, Q-Q plots)
- Homogeneity of Variance (Levene's test)
- Sphericity (Mauchly's test for repeated measures)

2 Nonparametric Alternatives

Use when assumptions are violated or data is ordinal (e.g., small sample Likert ratings).

- **Mann-Whitney U:** Alternative to Independent t-test
- **Wilcoxon Signed-Rank:** Alternative to Paired t-test
- **Kruskal-Wallis:** Alternative to One-way ANOVA
- **Friedman Test:** Alternative to Repeated Measures ANOVA

3 Transparent Reporting Guidelines

Move beyond *p*-values to provide complete statistical evidence for design decisions.

- Report exact p-values (e.g., $p = .023$, not $p < .05$) unless $p < .001$
- Always include effect sizes (Cohen's *d*, ηp^2) to show practical significance
- Display data distribution (boxplots, violin plots) rather than just bar charts with error bars

Integrating Across Study Types

Research types are not mutually exclusive; they form a development pipeline.

Phase 1: Scope

Descriptive Research

"Define the problem space"

Example Flow
Task Analysis & Field Observation

- Identify variables
- Understand context

Phase 2: Explain

Experimental Research

"Test causal mechanisms"

Example Flow
Factorial Experiment (2×2 Design)

- Compare alternatives
- Verify theories

Phase 3: Decide

Evaluation Research

"Validate the solution"

Example Flow
Usability Benchmarking

- Pass/Fail decisions
- Final validation

Iterative Refinement

Closing the Loop
Translate research findings into specific, actionable design requirements.

Input **User Needs** → Process **Evidence-Based Research** → Output **Better Systems**

Showcase: Motion Seat System

Open Access Article

Effects of a Motion Seat System on Driver's Passive Task-Related Fatigue: An On-Road Driving Study

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Abstract

Passive task-related (TR) fatigue caused by monotonous driving can negatively affect driving safety by impairing driver alertness and performance. This study aims to evaluate the effectiveness of a motion seat system on the driver's passive TR fatigue in terms of driving performance, physiological response, and subjective fatigue by using automotive and physiological sensors those applicable to on-road driving environment. Twenty drivers (5 females and 15 males; age = 38.5 ± 12.2) with more than two years of driving experience participated in an on-road experiment with two driving conditions: driving in the static seat condition during the first half of the driving session and then in the static (static–static, SS) or motion seat (static–motion, SM) condition during the second half. The SM condition showed significantly lower passive TR fatigue by 4.4~56.5% compared to the SS condition in terms of the standard deviation of velocity, percentage of eyelid closure rate (PERCLOS), and the ratio of low- to high-frequency power (LF/HF) of electrocardiography signals. The drivers rated significantly lower subjective state changes of overall fatigue, mental fatigue, passive TR fatigue, drowsiness, and decreased concentration in the SM condition than those in the SS condition. The findings of the study support the use of a motion seat system can be an effective countermeasure to reduce passive TR fatigue.

Keywords: passive task-related driver fatigue; monotonous driving; fatigue countermeasure; motion seat system

The Effects of a Motion Seat System on Driver's Passive Task-Related Fatigue: A Lab-Based Driving Simulation Study

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